

KANGGYE, Korea, Democratic People's Republic ofWMO: **470200**Lat: **40.967N**Lon: **126.600E**Elev: **306**StdP: **97.70**Time Zone: **9.00 (E09)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.2	-21.3	-25.4	0.4	-22.6	-23.5	0.5	-20.7	5.0	-7.0	4.1	-7.6	0.0	270	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	31.7	24.0	30.3	23.2	28.9	22.6	25.5	30.0	24.5	28.8	23.6	27.5	1.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.8	28.7	23.1	18.6	27.3	22.3	17.6	26.2	80.4	29.9	76.1	29.0	72.4	27.6	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.0	3.9	3.1	-26.2	34.0	2.2	1.7	-27.8	35.2	-29.1	36.2	-30.3	37.1	-31.9	38.3		
(5)			-26.4	27.4	2.1	1.7	-27.9	28.6	-29.2	29.6	-30.4	30.6	-31.9	31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-12.1	-7.3	1.0	9.2	15.9	20.8	23.9	23.2	17.2	9.5	0.1	-8.9		
	DBStd	12.99	4.61	5.11	4.34	4.28	3.37	2.69	2.22	2.55	3.21	4.02	5.26	4.90		
	HDD10.0	2457	684	483	281	64	2	0	0	0	1	58	298	586		
	HDD18.3	4295	943	717	538	274	89	9	0	2	58	274	547	844		
	CDD10.0	1649	0	0	1	41	184	324	431	409	217	42	1	0		
	CDD18.3	446	0	0	0	1	13	82	173	152	24	0	0	0		
	CDH23.3	3266	0	0	0	20	213	651	1219	1023	136	3	0	0		
	CDH26.7	956	0	0	0	1	44	193	393	309	14	0	0	0		
(14) Wind	WSAvg	0.5	0.3	0.4	0.7	0.9	0.8	0.5	0.4	0.4	0.4	0.4	0.5	0.3		
(15) Precipitation	PrecAvg	847	9	13	23	47	77	127	240	172	72	44	35	16		
	PrecMax	1276	35	53	56	101	201	289	543	379	227	82	105	44		
	PrecMin	603	1	2	1	8	20	35	111	68	5	1	5	4		
	PrecStd	191	8	13	15	22	41	66	108	72	51	23	21	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	8.1	16.5	26.0	30.2	32.6	34.4	33.7	28.4	23.6	14.4	5.8		
		MCWB	0.5	4.5	9.7	16.4	19.3	22.4	24.9	26.0	21.2	16.5	9.7	3.7		
	2%	DB	-0.3	4.9	12.9	22.8	27.8	30.6	31.9	31.3	26.5	20.7	11.7	2.4		
		MCWB	-2.0	1.9	7.3	13.9	18.1	21.6	24.6	24.7	20.0	14.4	7.8	0.9		
	5%	DB	-2.5	2.7	10.4	20.4	25.7	28.9	30.3	29.7	24.9	18.6	9.6	0.4		
		MCWB	-4.0	0.4	5.5	12.5	17.1	21.0	23.9	23.8	18.8	12.9	6.7	-0.9		
	10%	DB	-4.4	0.9	8.3	17.9	23.7	27.2	28.8	28.3	23.3	16.5	7.5	-1.3		
		MCWB	-5.7	-1.0	4.3	11.1	16.0	20.3	23.4	22.9	17.9	11.7	5.3	-2.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	4.8	10.3	18.0	21.3	24.2	26.9	27.2	22.7	17.3	10.5	3.8		
		MCDB	2.2	7.5	15.2	24.5	28.3	29.9	31.8	31.8	26.8	22.3	14.2	5.3		
	2%	WB	-1.8	2.3	7.6	15.1	19.3	23.0	25.6	25.6	21.1	15.2	8.5	0.9		
		MCDB	-0.4	4.3	12.2	21.2	25.7	28.9	30.3	29.9	25.0	19.4	10.9	2.1		
	5%	WB	-3.8	0.6	6.0	13.1	18.0	21.9	24.7	24.6	19.9	13.6	7.0	-0.9		
		MCDB	-2.5	2.5	9.9	19.3	24.3	27.4	29.1	28.7	23.4	17.7	9.3	0.2		
	10%	WB	-5.6	-0.9	4.5	11.4	16.7	20.9	23.9	23.6	18.8	12.2	5.3	-2.5		
		MCDB	-4.4	0.9	8.0	16.9	22.2	26.0	27.8	27.3	22.3	15.7	7.3	-1.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.8	10.6	12.1	12.4	10.8	8.6	9.0	10.9	11.8	9.0	9.9		
		MCDBR	11.7	11.8	13.9	17.0	16.7	14.4	11.5	11.1	12.4	14.9	11.8	9.8		
		MCWBR	10.1	9.2	9.0	9.5	8.5	7.1	5.9	5.8	6.7	8.7	8.3	8.4		
		MCDBR	11.4	10.9	12.8	15.2	14.5	12.2	9.7	9.7	10.6	13.2	10.8	9.5		
	5% WB	MCWBR	10.0	8.7	8.9	9.3	8.2	6.8	5.8	5.8	6.5	8.4	8.0	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.385	0.468	0.552	0.627	0.661	0.706	0.640	0.561	0.495	0.522	0.454	0.396		
	taud		1.913	1.753	1.651	1.585	1.569	1.553	1.733	1.906	1.990	1.825	1.887	1.932		
	Ebn at Noon		709	697	687	673	665	636	676	719	737	640	625	655		
	Edh at Noon		132	182	226	258	269	273	226	185	159	168	137	121		
(44) All-Sky Solar Radiation	RadAvg		2.50	3.38	4.37	4.81	5.40	5.09	4.46	4.51	4.32	3.31	2.17	2.01		
	RadStd		0.13	0.22	0.26	0.36	0.46	0.53	0.52	0.49	0.41	0.25	0.26	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.75	+0.79	N/A	N/A	+72	N/A
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page